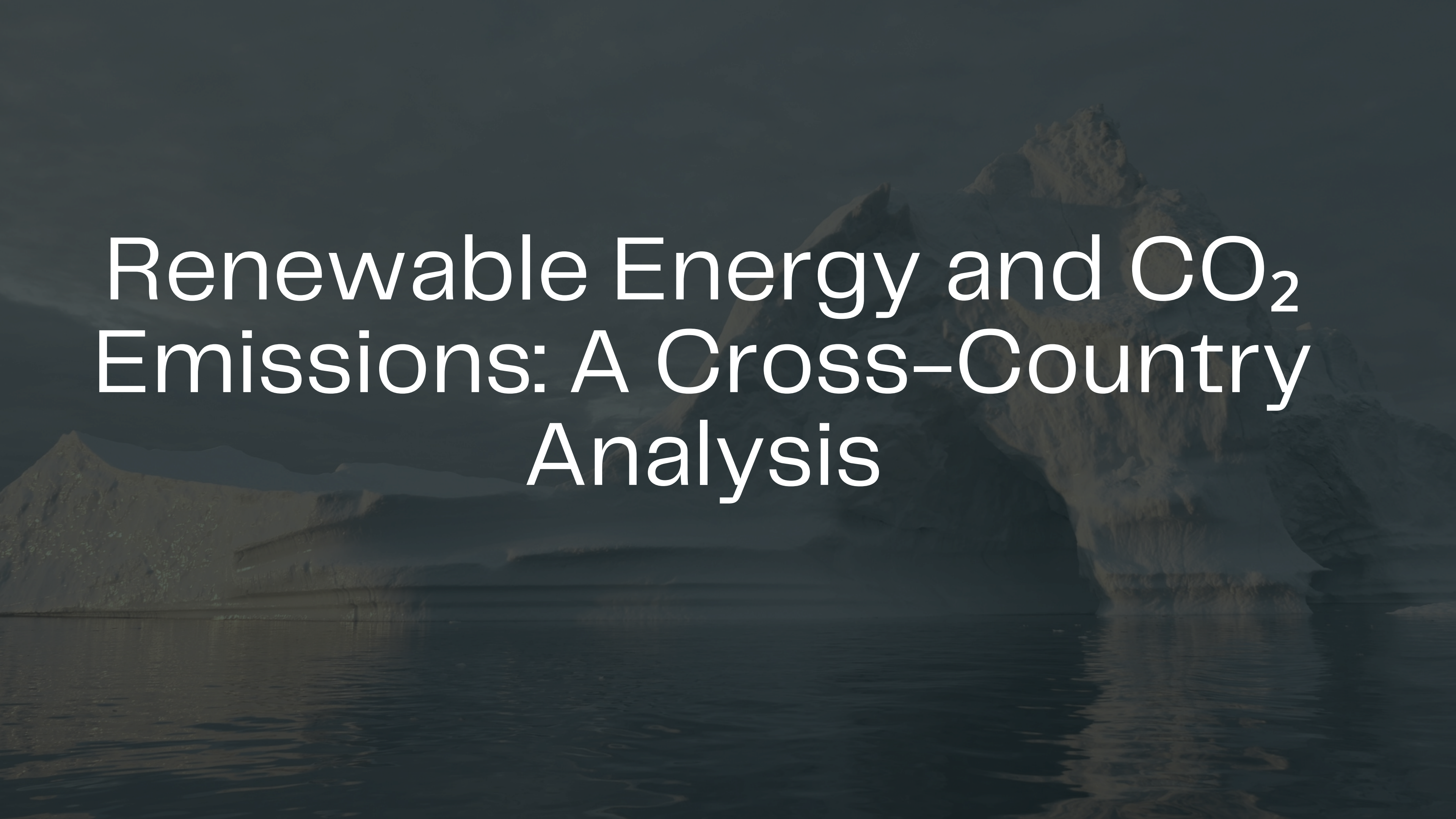


A large, dark iceberg floats in a calm body of water under a dark, overcast sky. The iceberg's jagged peaks and ridges are visible, and its reflection is clearly seen in the still water below. The overall mood is somber and contemplative.

Renewable Energy and CO₂ Emissions: A Cross-Country Analysis

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ABSTRACT

This project investigates the relationship between renewable energy usage and CO₂ emissions across different countries. The study uses publicly available data from Our World in Data and focuses on seven countries: Germany, France, Italy, Spain, China, India, and the United States.

The collected data was cleaned, organized, and stored in a SQLite database. SQL queries were used to analyze the data, while data visualizations helped identify trends and patterns. A Linear Regression model was also developed to explore whether renewable energy usage can help explain changes in CO₂ emissions.

The results indicate that countries with higher renewable energy adoption generally tend to have lower CO₂ emissions, although renewable energy share alone is not sufficient to accurately predict emission levels. The project demonstrates how data science techniques can be applied to sustainability-related problems and provides insights into global energy transition efforts.

INTRODUCTION

Climate change has become one of the most important global challenges of the 21st century. Increasing greenhouse gas emissions, especially carbon dioxide (CO₂), have contributed significantly to global warming and environmental degradation.

Many countries are investing in renewable energy sources such as solar, wind, and hydropower to reduce their dependence on fossil fuels. Renewable energy is considered a key solution for reducing carbon emissions and creating a more sustainable future.

This project aims to examine whether countries that use a higher share of renewable energy also tend to produce lower CO₂ emissions. By analyzing real-world data from multiple countries over several years, the study seeks to understand the relationship between energy transition and environmental sustainability.



RESEARCH QUESTION AND OBJECTIVES



Research Question

What is the relationship between renewable energy use and CO₂ emissions across different countries?

Objectives

- Compare renewable energy usage among selected countries.
- Compare CO₂ emissions across different countries.
- Analyze trends in renewable energy adoption over time.
- Examine possible relationships between renewable energy usage and CO₂ emissions.
- Apply data science techniques such as SQL analysis, visualization, and machine learning to extract insights from the data.

DATASET DESCRIPTION

The data used in this project was obtained from publicly available datasets provided by Our World in Data (OWID). Two datasets were used:

Energy Dataset

The energy dataset contains information about energy production, energy consumption, and renewable energy usage for countries around the world.

Important variables used:

- Country
- Year
- Renewables Share Energy (%)

CO₂ Dataset

The CO₂ dataset contains information about carbon dioxide emissions for countries worldwide.

Important variables used:

- Country
- Year
- CO₂ Emissions

Selected Countries

To keep the analysis manageable and meaningful, seven countries were selected:

- Germany
- France
- Italy
- Spain
- China
- India
- United States

The study covers the years 2000–2024.

DATA COLLECTION AND CLEANING

The raw datasets contained hundreds of variables and thousands of records. Therefore, several preprocessing steps were performed before analysis.

Data Cleaning Steps

1. Loaded both datasets using the Pandas library.
2. Selected only the variables relevant to the project.
3. Filtered the data to include the selected countries.
4. Removed unnecessary columns.
5. Merged the renewable energy dataset with the CO₂ emissions dataset using Country and Year as common keys.
6. Checked for missing values and removed incomplete records where necessary.

After cleaning, a final dataset was created containing:

- Country
- Year
- Renewable Energy Share
- CO₂ Emissions

This cleaned dataset was used for database creation, visualization, and machine learning.

Database Design

A SQLite database was created to store and organize the cleaned sustainability data.

The database consists of three tables:

- Countries
 - Stores basic country information.
 - Fields:
 - country_id (Primary Key)
 - country_name
- RenewableEnergy
 - Stores renewable energy share data.
 - Fields:
 - energy_id (Primary Key)
 - country_id (Foreign Key)
 - year
 - renewables_share_energy
- CO2Emissions
 - Stores CO₂ emission data.
 - Fields:
 - emission_id (Primary Key)
 - country_id (Foreign Key)
 - year
 - co2

SQL ANALYSIS

Three SQL queries were developed to analyze the dataset.

Query 1: Average Renewable Energy Share by Country

This query calculates the average renewable energy share for each country.

Finding:

Spain, Italy, and Germany showed the highest average renewable energy usage among the selected countries.

Query 2: Average CO₂ Emissions by Country

This query calculates the average CO₂ emissions for each country.

Finding:

China and the United States recorded the highest average CO₂ emissions, while Spain showed comparatively lower emission levels.

Query 3: Combined Renewable Energy and CO₂ Data

This query joins all three tables and displays renewable energy share and CO₂ emissions for each country and year.

Finding:

The combined dataset provides a clear view of sustainability trends over time and serves as the foundation for visualization and machine learning analysis.

VISUALIZATION 1

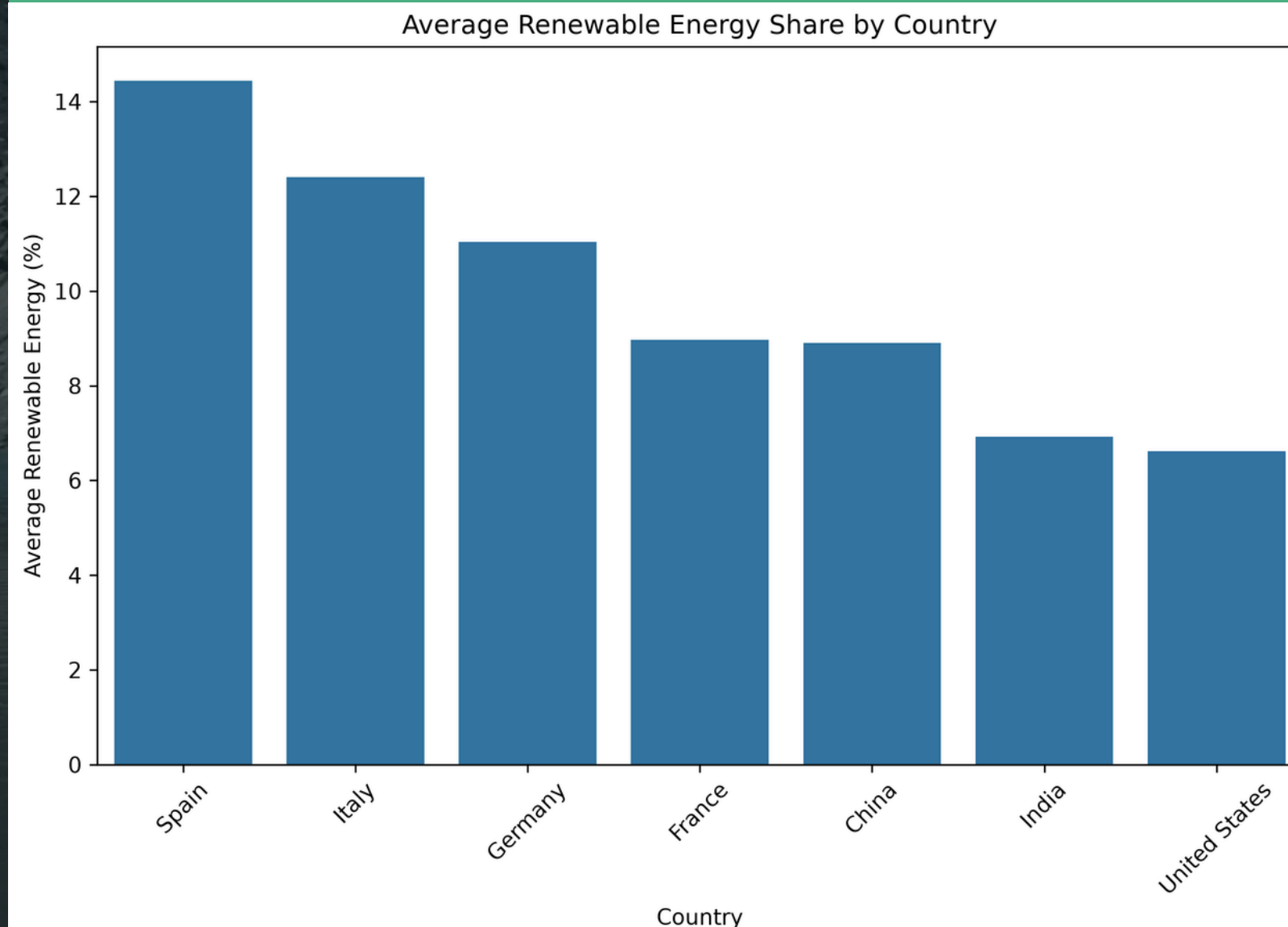
Analysis of Renewable Energy Share

The first visualization compares the average renewable energy share among the selected countries.

The results show that Spain has the highest average renewable energy share, followed by Italy and Germany. These countries have made significant investments in renewable energy sources such as wind and solar power over the past two decades.

China, India, and the United States show comparatively lower renewable energy shares despite their large investments in renewable technologies. This is mainly due to their overall energy demand and continued dependence on fossil fuels.

The graph highlights how renewable energy adoption varies significantly across countries and demonstrates different stages of energy transition.



VISUALIZATION 2

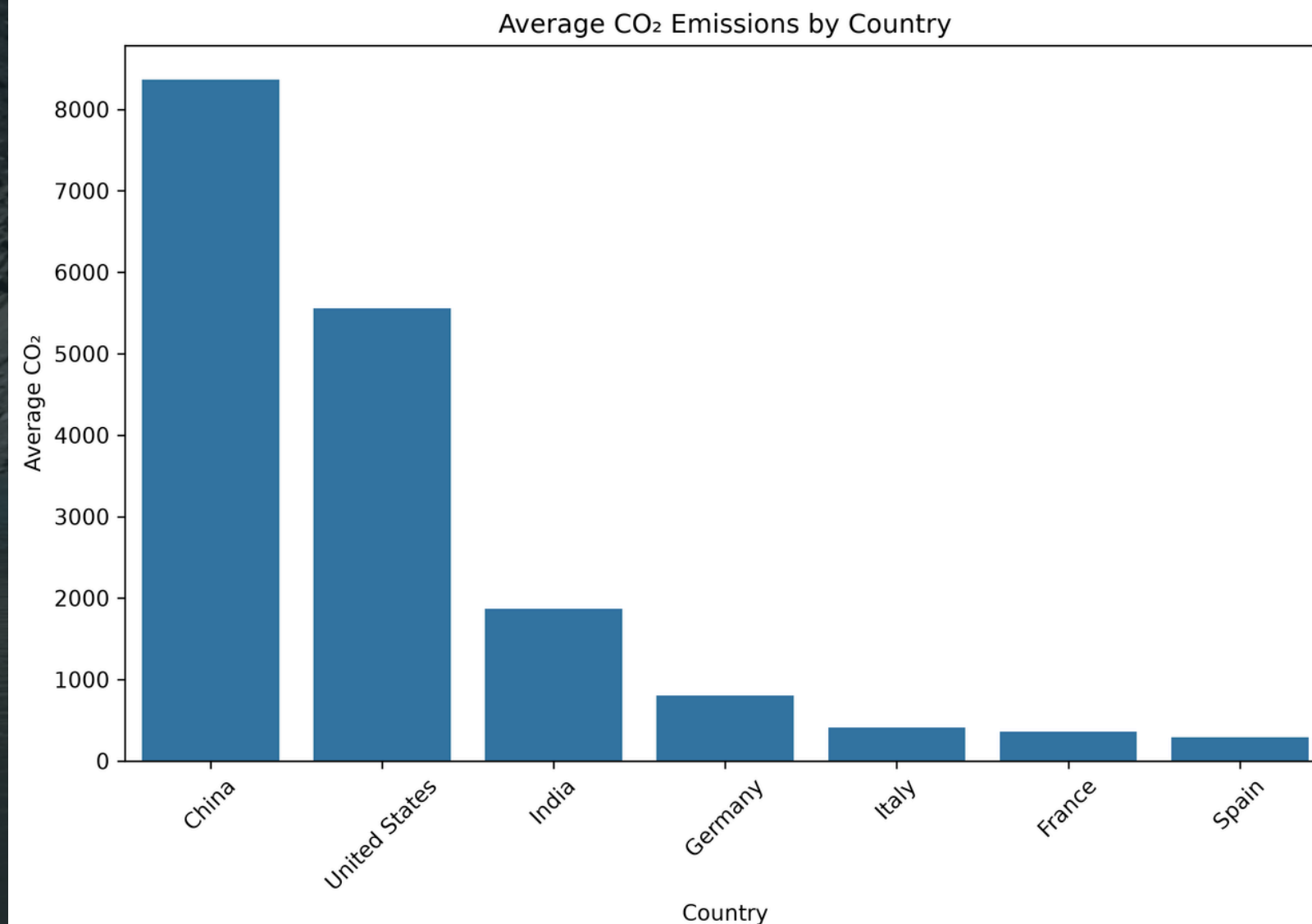
Analysis of CO₂ Emissions

The second visualization compares the average CO₂ emissions of the selected countries.

China recorded the highest average emissions among all countries analyzed. The United States also showed significantly high emission levels. Both countries have large industrial sectors and high energy consumption. European countries such as Spain, France, Germany, and Italy exhibited lower average emissions compared to China and the United States.

India's emissions have increased steadily due to economic growth and industrial development, although they remain lower than those of China and the United States.

This visualization provides a clear understanding of how emission levels differ across developed and developing economies.



VISUALIZATION 3

Relationship Between Renewable Energy and CO₂ Emissions

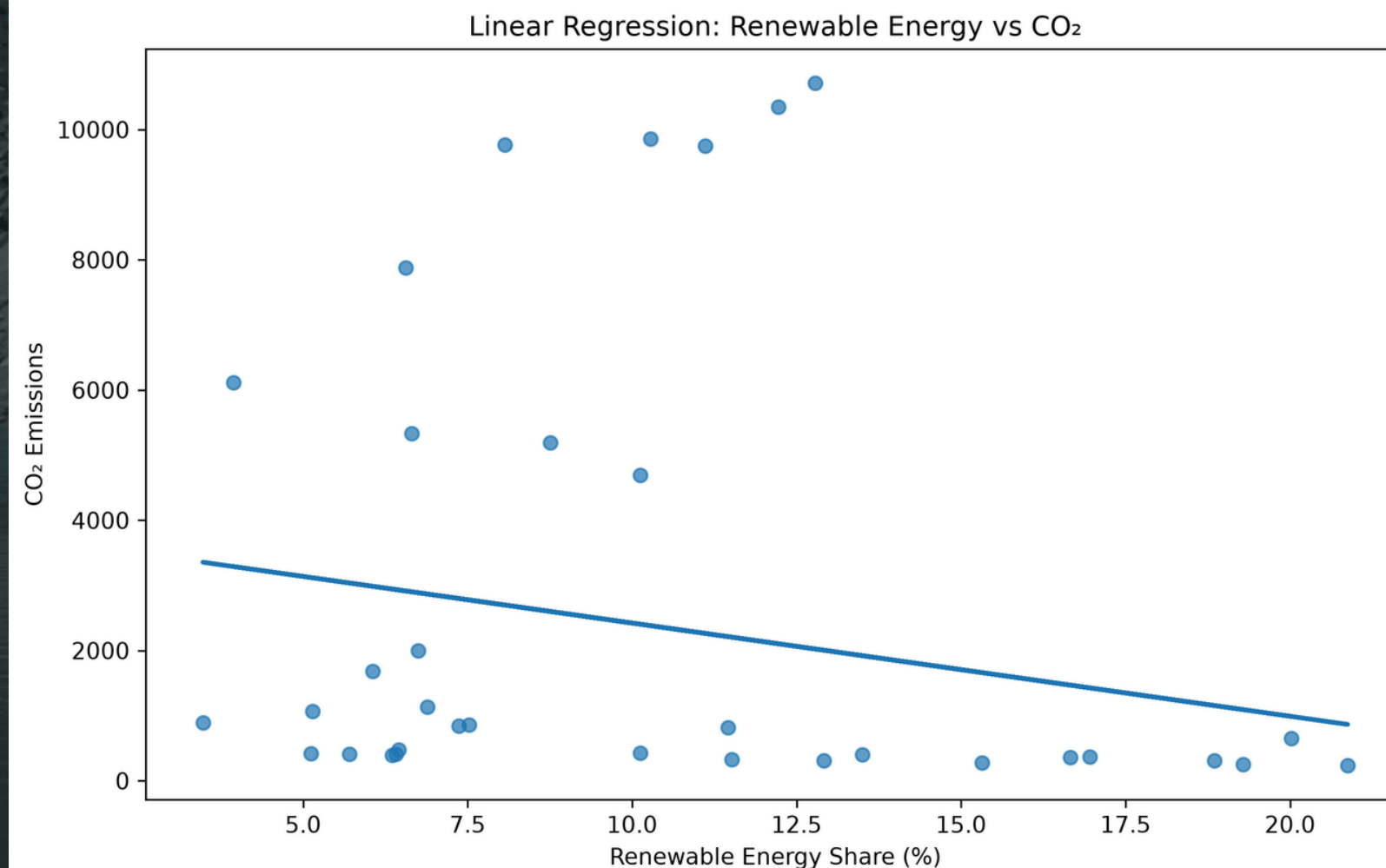
To explore the relationship between renewable energy adoption and CO₂ emissions, a scatter plot with a linear regression line was created.

The regression line shows a slight downward trend, suggesting a negative relationship between renewable energy share and CO₂ emissions. In general, countries with higher renewable energy usage tend to produce lower emissions.

However, the data points are widely spread, indicating that renewable energy alone does not fully explain emission levels.

Other factors such as population size, industrial activity, GDP, energy demand, and government policies also influence carbon emissions.

The scatter plot provides an important visual representation of the complexity of sustainability-related issues.



MACHINE LEARNING MODEL

Linear Regression Model

A Linear Regression model was developed to investigate whether renewable energy share can be used to predict CO₂ emissions.

The model used:

Input Variable (X):

- Renewable Energy Share (%)

Target Variable (Y):

- CO₂ Emissions

The dataset was divided into training and testing sets.

The model was trained using historical data and evaluated using common regression metrics

Model Performance

MAE (Mean Absolute Error): 2762.73

MSE (Mean Squared Error): 12,736,306.19

R² Score: 0.0062

Interpretation

The R² score is very close to zero, which indicates that renewable energy share alone has very limited predictive power for CO₂ emissions.

This result is reasonable because emissions are influenced by many additional variables, including:

- Population
- Economic growth
- Industrial activity
- Energy consumption
- Government regulations
-

Although the predictive performance is weak, the model successfully demonstrates the application of machine learning techniques to sustainability data.

RESULTS AND FINDINGS

Key Findings

The analysis produced several important findings:

- Spain achieved the highest average renewable energy share among the selected countries.
- China recorded the highest average CO₂ emissions.
- Germany demonstrated a consistent increase in renewable energy adoption over recent years.
- Countries with higher renewable energy usage generally tended to have lower emissions.
- The relationship between renewable energy and emissions exists but is relatively weak when analyzed using a single variable.
- Multiple economic and environmental factors influence carbon emissions.

Overall, the findings support the importance of renewable energy while highlighting the complexity of climate-related challenges.

Limitations and Future Improvements

Limitations

This project has several limitations.

First, only seven countries were included in the analysis. A larger dataset could provide more reliable results.

Second, the machine learning model used only one predictor variable, renewable energy share.

Third, some missing values were present in the original datasets and had to be excluded during cleaning.

Future Improvements

Future work could include:

- Adding more countries.
- Including GDP and population data.
- Using advanced machine learning algorithms.
- Analyzing additional sustainability indicators.
- Developing an interactive dashboard using Power BI or Tableau.

These improvements could provide deeper insights into global sustainability trends.

CONCLUSION

The objective of this project was to investigate the relationship between renewable energy usage and CO₂ emissions across different countries.

Using publicly available datasets, a SQLite database was created, SQL analysis was performed, multiple visualizations were generated, and a machine learning model was developed.

The results indicate that renewable energy adoption is generally associated with lower emissions. However, renewable energy alone is not sufficient to explain emission levels, as many economic and social factors also play important roles. The project demonstrates how data science techniques can be used to analyze sustainability challenges and support evidence-based decision making.

Project Reflection

Reflection

This project provided practical experience in applying data science concepts to a real-world sustainability problem.

Throughout the project, I learned how to collect data from public sources, clean and prepare datasets, design a relational database, write SQL queries, create visualizations, and develop a machine learning model.

One of the biggest challenges was combining data from multiple datasets and ensuring consistency between renewable energy and CO₂ emission records. This process improved my understanding of data preprocessing and database management.

The project also demonstrated that real-world environmental issues are complex and cannot always be explained by a single variable. While renewable energy adoption is important, many other factors influence carbon emissions.

Overall, this project strengthened my skills in data analysis, database design, visualization, and machine learning while providing valuable insights into sustainability and environmental data.